

CPU Scheduling Exercises 6.16–6.19

6.16 Consider the following set of processes, with the length of the CPU burst given in milliseconds:

Process	Burst Time	Priority
P ₁	2	2
P ₂	1	1
P ₃	8	4
P ₄	4	2
P ₅	5	3

The processes are assumed to have arrived in the order P₁, P₂, P₃, P₄, P₅, all at time 0.

a. Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, nonpreemptive priority (a larger priority number implies a higher priority), and RR (quantum = 2).

6.17 The following processes are being scheduled using a preemptive, round-robin scheduling algorithm. Each process is assigned a numerical priority, with a higher number indicating a higher relative priority. In addition to the processes listed below, the system also has an idle task (which consumes no CPU resources and is identified as P_{idle}). This task has priority 0 and is scheduled whenever the system has no other available processes to run. The length of a time quantum is 10 units. If a process is preempted by a higher-priority process, the preempted process is placed at the end of the queue.

Thread	Priority	Burst	Arrival
P ₁	40	20	0
P ₂	30	25	25
P ₃	30	25	30
P ₄	35	15	60
P ₅	5	10	100
P ₆	10	10	105

a. Show the scheduling order of the processes using a Gantt chart.

b. What is the turnaround time for each process?

c. What is the waiting time for each process?

d. What is the CPU utilization rate?

6.19 Which of the following scheduling algorithms could result in starvation? Select all that apply.

- a. First-come, first-served
- b. Shortest job first
- c. Round robin
- d. Priority